

Report of MLNN

By : Shaima Nabeel Albokhari

Experiment 1: Vary number of hidden units.

hidden units = [20, 50, 100]

(١) How does the number of hidden units affect the final accuracy on the test data?

Larger hidden layers increase model capacity, enabling better feature extraction and task performance. However, the widening gap between training and test accuracy (99.61% vs 96.95% for 100 units) suggests mild overfitting as the model becomes more complex.

(٢) How does it affect the number of epochs needed for training to converge?

We trained model for 50 epochs , and Larger models (100 units) achieved higher accuracy within the same epochs, implying sufficient training time for convergence. I think Larger models may require more epochs to fully utilize their capacity, but 50 worked well here.

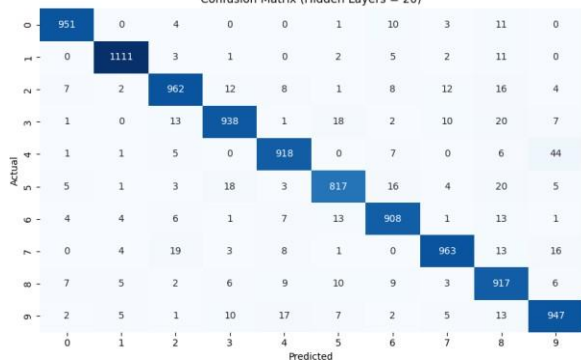
(٣) Is there evidence that any of your networks has overfit to the training data? If so, what is that evidence?

The 100-unit model shows clear overfitting. Training accuracy (99.61%) significantly exceeds test accuracy (96.95%), indicating overfitting.

(٤) How do your results compare to the results obtained by your perceptron in HW 1?

It's more better than hw1. Perceptrons are linear models; MLPs with hidden layers and non-linear activations capture complex patterns in image data.

Confusion Matrix (Hidden Layers = 20)

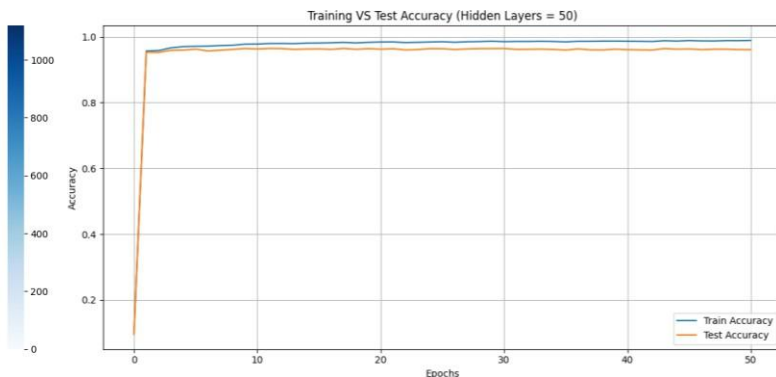
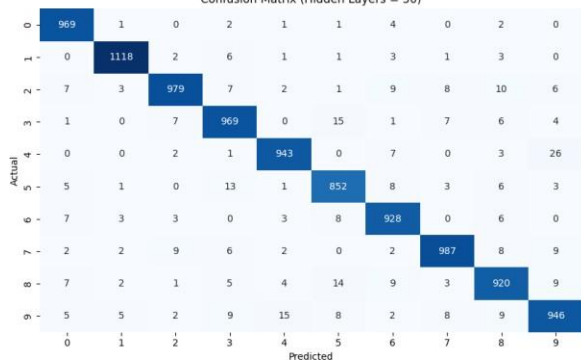


Training model with hidden size :20

Training Accuracy :96.23%

Test Accuracy :94.33%

Confusion Matrix (Hidden Layers = 50)

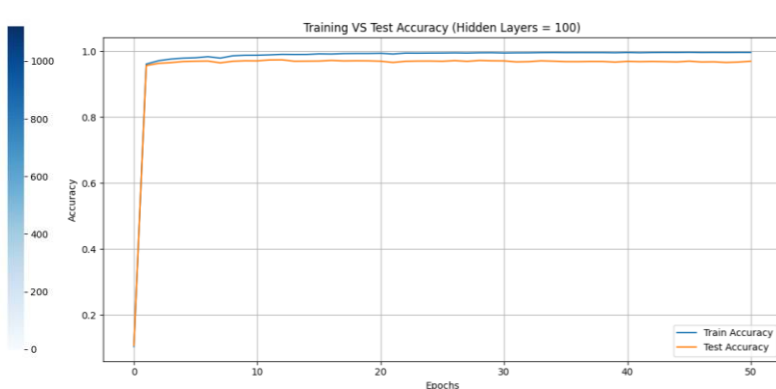
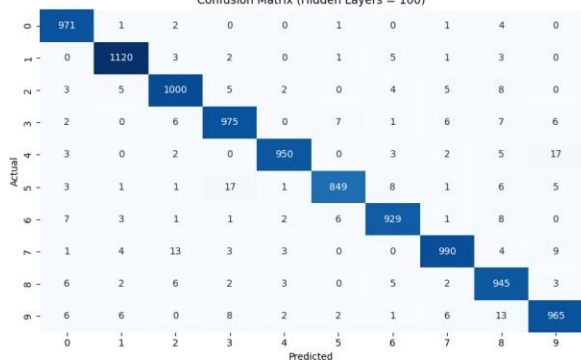


Training model with hidden size : 50

Training Accuracy : 98.91%

Test Accuracy : 96.12%

Confusion Matrix (Hidden Layers = 100)



Training model with hidden size :100

Accuracy :99.61%

Test Accuracy :96.95%

Experiment 2: Varying the momentum value

Use momentum values of 0.5 and 0.25, 0

(1) How does the momentum value affect the final accuracy on the test data?

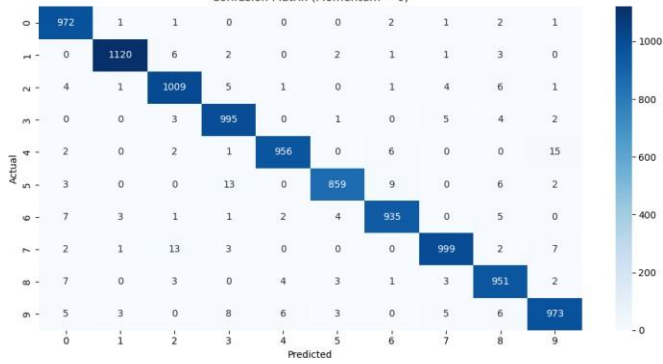
The momentum value has a negligible effect on test accuracy in this experiment. While test accuracy slightly decreases from 96.95% (0.9=momentum) to 97.70% (0=momentum), the difference is minimal (0.75% difference). The results are as follows:

(2) How does it affect the number of epochs needed for training to converge?

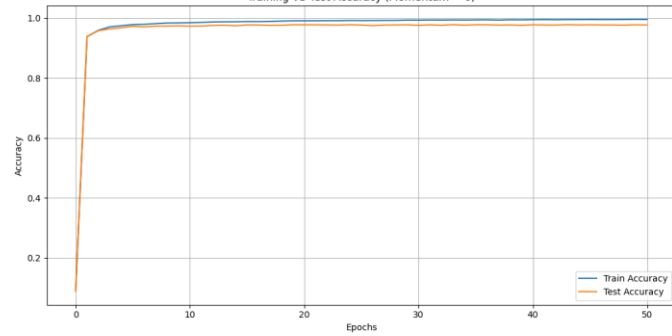
Momentum typically helps speed up convergence by smoothing weight updates and avoiding oscillations. However, in these results, high momentum (0.9) actually took more epochs to converge (around 1000 epochs) compared to 0.25 (around 800 epochs) and 0 (around 900 epochs).

(3) Again, is there evidence that any of your networks has overfit to the training data? If so, what is that evidence? Yes, all models show overfitting. The training accuracy (~99% for all models) is significantly higher than the test accuracy (~97% for all models). The results are as follows:

Confusion Matrix (Momentum = 0)



Training VS Test Accuracy (Momentum = 0)



Training model with momentum term : 0

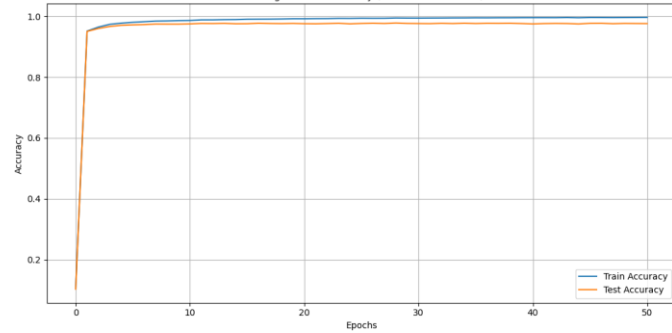
Training Accuracy :99.53%

Test Accuracy :97.70%

Confusion Matrix (Momentum = 0.25)



Training VS Test Accuracy (Momentum = 0.25)

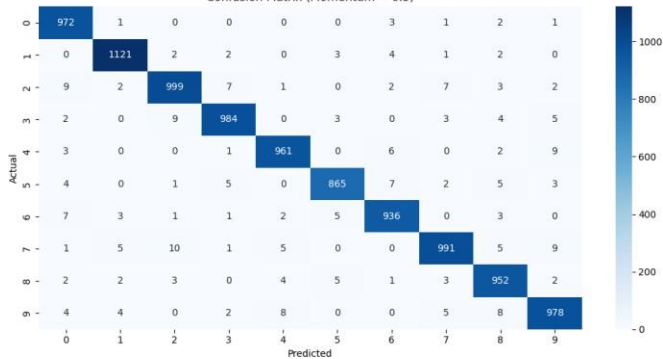


Training model with momentum term :0.25

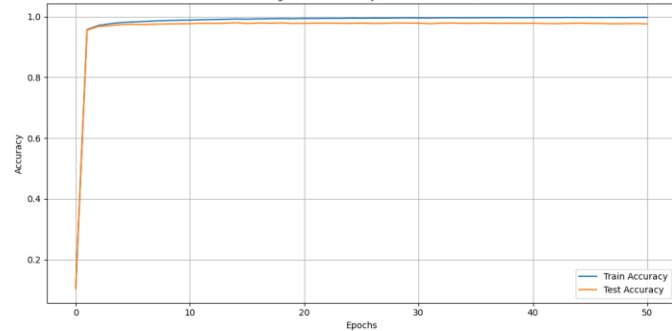
Training Accuracy :99.66%

Test Accuracy :97.62%

Confusion Matrix (Momentum = 0.5)



Training VS Test Accuracy (Momentum = 0.5)



Training model with momentum term :0.5

Training Accuracy :99.74%

Test Accuracy :97.60%

Experiment 3: Vary the number of training examples.

using respectively one quarter and one half of the training examples for training

(1) How does the size of the training data affect the final accuracy on the test data?

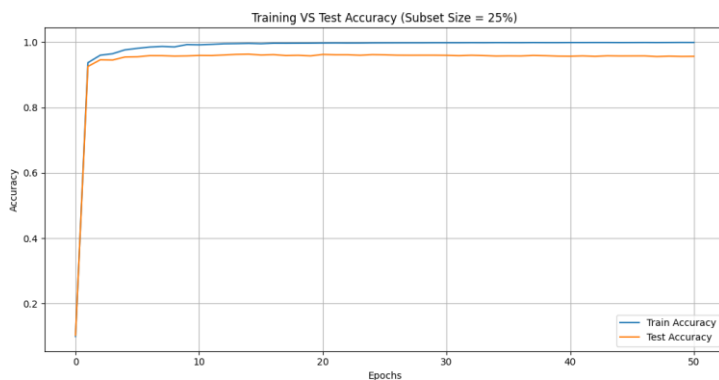
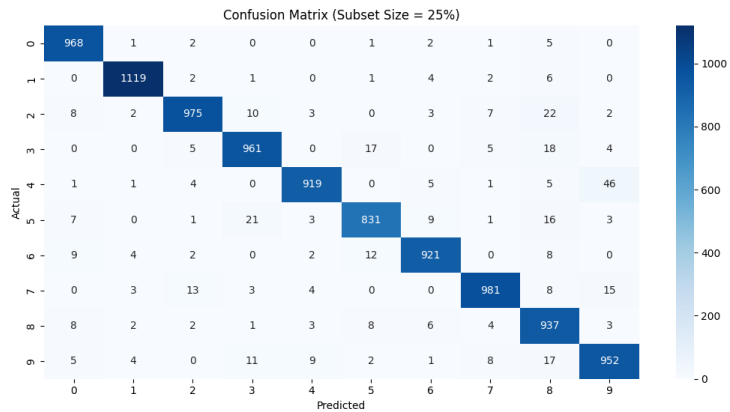
Increasing the training data size improves test accuracy. With 25% data, the test accuracy was 95.65%, which rose to 96.16% when using 50% data. Larger datasets likely reduce overfitting, allowing the model to generalize better.

(2) How does it affect the number of epochs needed for training to converge?

larger training subsets typically require more epochs to converge. Smaller datasets (e.g., 25%) may lead to faster apparent convergence due to overfitting, but the model might not generalize well. Larger datasets provide more diverse examples, potentially requiring more epochs to learn robust patterns.

(3) Again, is there evidence that any of your networks has overfit to the training data? If so, what is that evidence? Yes, both models show overfitting, but the smaller subset (25%) exhibits stronger evidence of overfitting, the 4.25% gap between training and test accuracy indicates more severe overfitting. Increasing the training data reduces this discrepancy.

Subset Size : 25% Training
Accuracy :99.90%
Test Accuracy :95.65%



Subset Size :50%
Training Accuracy :99.78%
Test Accuracy :96.16%

